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Video Capture and Output of the G2 Amiga

The Amiga's native picture captured at 640 samples a line into three frame buffers in DDR2 and shown on both G2 video heads, scaled to 640 × 480 at 31 kHz for any VGA, DVI or HDMI monitor.

HIGHLIGHTS

- 640 samples a line, RGB565, triple-buffered in DDR2.
- Two heads: DVI-I (with analogue VGA) and micro-HDMI, one or both.
- NTSC and PAL; interlaced screens shown in full.
- Picture offset adjustable from setup.

KEY PARAMETERS

PARAMETER	VALUE
Samples a line	640
Format	RGB565
Frame buffers	3 × 1 MB
Output	640 × 480, 31.5 kHz
Heads	2 × CH7301C
Default offset	77 samples

1 PICTURE PATH

STAGE	IMPLEMENTATION
Source	The core's native picture before its scan doubler, with the OSD layer, RGB
Sampling	Every second core edge on 15 kHz lines (14 MHz grid: hires pixels once, lowres twice), every core edge on 31 kHz lines; the line rate measured per line
Window	640 samples a line from sample HOFS after horizontal blank (default 77, the standard display window); outside vertical blank
Format	RGB565, a two-line buffer, each finished line written to DDR2 chip B
Frame buffers	Three of 1 MB in chip B, written in turn; a mirror 16 MB higher for the second head; interlaced frames place line y at 2y + field (a 484-line picture)
Heads	Two Chrontel CH7301C encoders: DVI-I (digital and analogue VGA) and micro-HDMI; each scans the last finished buffer

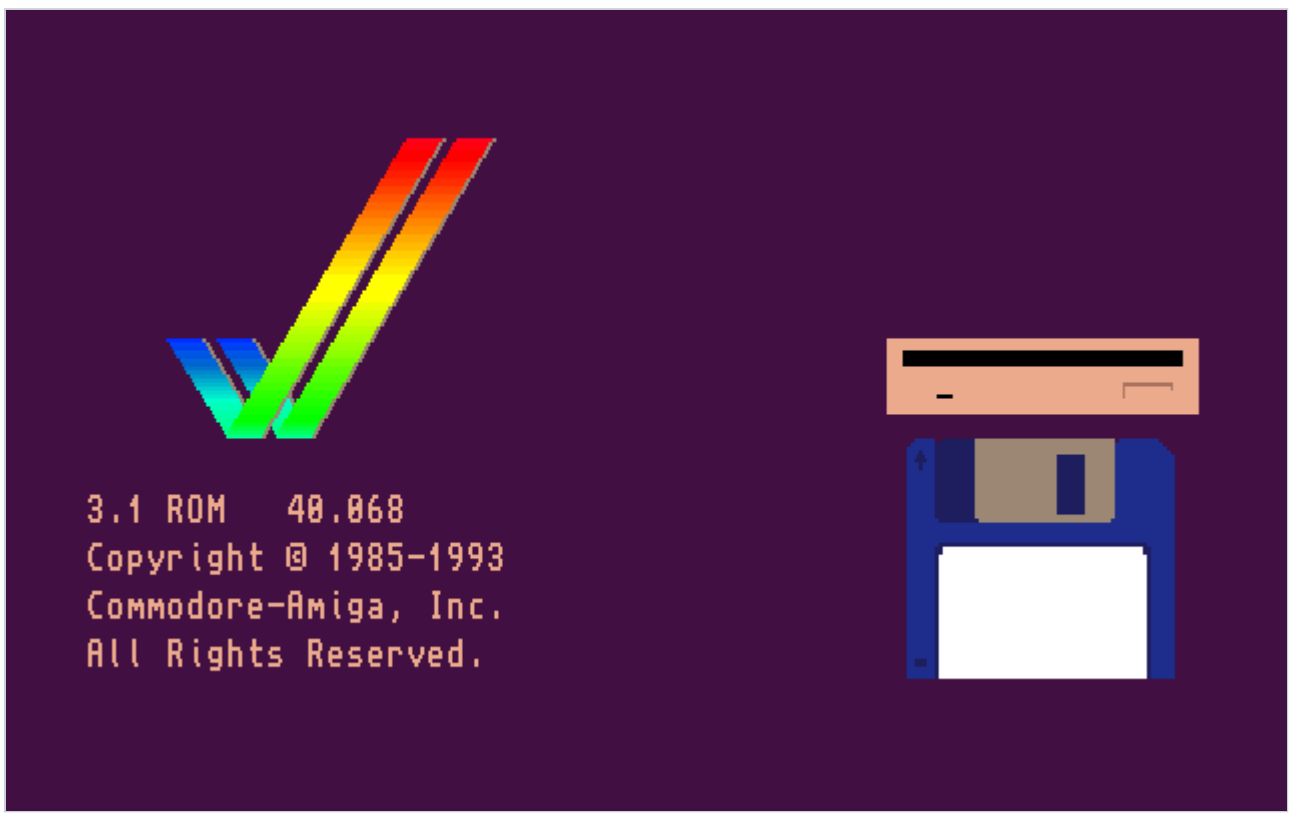


Figure 1. A captured frame of the Kickstart 3.1 (40.068) insert-disk screen: the NTSC display window, 640 samples × 200 lines from capture sample 77 after hblank (HOFS=77), each line shown twice.

2 OUTPUT MODES

VIDEO=	REGION	HEAD MODE	LINE RATE	USE
SCALE (default)	NTSC or PAL	640 × 480, 60 Hz	31.5 kHz	Any VGA, DVI or HDMI monitor: every Amiga mode scaled to it
NATIVE	NTSC	640 × 240, 59.72 Hz	15.7 kHz	A 15 kHz monitor (Amiga 1080, a TV through an adapter): untested on hardware
NATIVE	PAL	640 × 288, 50.15 Hz	15.6 kHz	As above: untested on hardware

KEY	VALUES	EFFECT
`SCREEN=`	0 micro-HDMI, 1 DVI-I, 2 both (default)	The heads that show the Amiga
`HOFS=`	0–400, default 77	First captured sample of a 15 kHz line; the console command `vid hofs` sets it at run time
`REGION=`	NTSC (default), PAL	Chipset clock and frame rate; the heads' source height 240 or 288 lines

3 LIMITS AND UNTESTED ITEMS

ITEM	STATE
31 kHz Amiga modes (Productivity, DblPAL, DblNTSC, Multiscan)	Captured line by line in simulation (0 mismatches in 185,856 pixels); untested on hardware
NATIVE output	Simulated; untested on hardware (needs a 15 kHz monitor)
A PAL screen on an NTSC-configured unit	The path shows 240 lines: the last 16 lines of a 256-line screen are not seen. Set REGION=PAL
RTG (graphics card) modes	Not present
Lines dropped	0 in normal operation (status counter LINE_DROPS)

Measured on a G2 revision C unit with firmware release 1.0 and fabric 0x0AB5EF2D unless stated. Repository mikewolak/pano-g2-amiga, release 1.0.